# DANAHER CORP DHR

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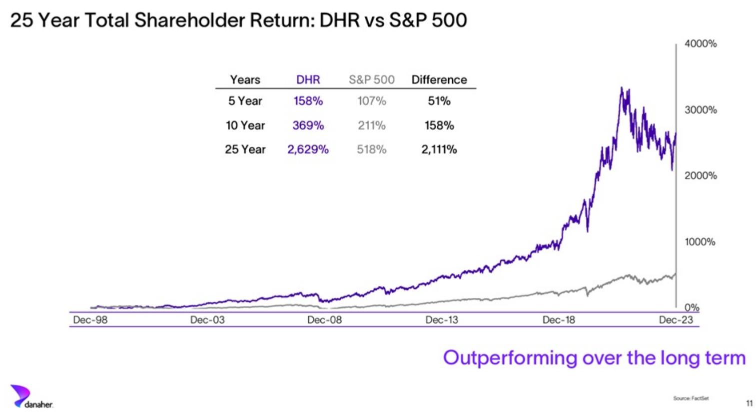
by valueinvestor03

| | | | 2024 | 2025 |
|---|---|---|---|---|
| Price: | 250.00 | EPS | 7.60 | 0 |
| Shares Out. (in M): | 741 | P/E | 32.9 | 0 |
| Market Cap (in $M): | 185,200 | P/FCF | 32 | 0 |
| Net Debt (in $M): | 1,110 | EBIT | 6,960 | 0 |
| TEV (in $M): | 196,300 | TEV/EBIT | 28 | 0 |

## Description

Introduction

Danaher (DHR) has produced one of the better long-term track records in the S&P 500 over the past few decades. Over the past 10 years, the stock price has compounded at 15%, despite still trading below its high of 2021. While the stock isn't cheap today, the price is fair for a very high-quality company which can compound free cash flow per share at DD rates.




Danaher was founded by Steven and Mitchell Rales in the early 1980's when the brothers started buying industrial companies of middling quality. In the mid-1980's, the name of the company was changed to Danaher (named after a fishing stream in Montana) and the company continued acquiring industrial assets with a strategy like private equity: paying low prices, financing with large amounts of junk bonds, and improving operations. However, unlike private equity, Danaher did not sell the companies it acquired, rather it paid down debt and rolled the cash flows into additional deals. A key deal came in 1986 when the company acquired Chicago Pneumatic, which was a conglomerate itself and owned a business called Jacobs Engine Brake. In a huge stroke of luck, having spent time working in manufacturing in Japan, one of the key managers at Jacobs had developed a fascination with Japanese manufacturing techniques such as Lean and Kaizen. Through the efforts of this manager, the Jacobs Engine Brake factory began to see significant improvements in cost, quality, productivity, and customer satisfaction. These results caught the attention of the Rales brothers who sought to document these methods and spread them throughout the other businesses within Danaher. The initial meetings led to the establishment of the Danaher Production System, which is now referred to as the Danaher Business System (DBS).

Today Danaher defines DBS as "a set of business processes and tools" that the company's operating businesses use to continuously improve over time. The five core values of DBS are:

- The Best Team Wins,
- Customers Talk, We Listen,
- Kaizen (continuous improvement) is our Way of Life,
- Innovation Defines our Future, and
- We Compete for Shareholders

These core values may seem like buzz words or empty slogans, but from everything I've read, plus the results the company has achieved over time, DBS is deeply engrained in Danaher's culture and drives everything the company does. The author of the book *Lessons From the Titans*, who covered Danaher on the sell-side for many years, wrote a chapter on Danaher and claimed the company may have "the most process-driven culture of any company in America."

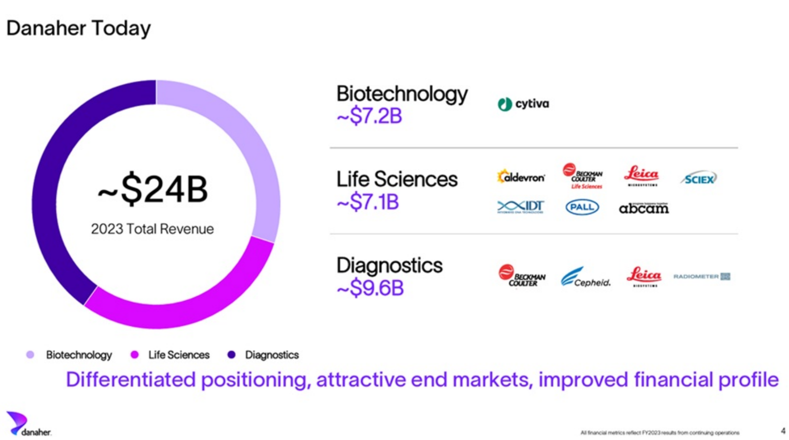
Throughout the late 1980s and 1990s, Danaher continued to acquire additional industrial B2B companies. It was also expanding the use of DBS from the factory floor to other areas including procurement and customer-facing functions. Over time the company's acquisition criteria evolved to include a greater focus on quality and stability as opposed to the cyclical nature of many of its earlier industrial deals. The company realized that DBS and the idea of continuous improvement works best with companies generating relatively steady growth (not too much, too cyclical, or worse, businesses in decline). Through DBS's productivity tools, manufacturing capacity could be increased by 2% to 3% annually without adding people or equipment. So, with relatively small amounts of capital expenditures, growth could be funded at a very low cost and the businesses would be capital efficient, thus freeing up more capital to invest in more acquisitions. However, if growth is too fast, the productivity tools can't keep up and capital intensity increases. Over time, the company came to appreciate the fact that DBS worked best with non-cyclical, high gross margin businesses, especially where there was a wide gap between gross and operating margins. A high gross margin was a signal that the customer valued the product and a wide gap between gross and operating margins meant there was potential for DBS to take out costs.

After the Hach acquisition in 1999, the Videojet acquisition in 2002, and deals in the healthcare industry later on, Danaher began to fully realize the immense benefits of selling consumables on a large installed base of equipment. For example, Videojet sold industrial printers which required a steady stream of ink sales. The company started looking to buy higher quality assets that possessed these characteristics which led the company into dental, niche electrical testing equipment, and life sciences. The life science platform began with the with the 2005 acquisition of Leica Microsystems, and since then, Danaher has deployed over $60 billion to acquire over 40 companies within that space. All the while, Danaher has continued to push DBS deeper into additional areas of the company including sales funnel management, value engineering, voice of the customer, value pricing, and procurement and logistics tools as necessitated by the fact that the company was moving away from its cyclical industrial roots. For example, a DBS tool might focus on reducing the number of parts in a product or standardizing the use of certain parts across more than one product. From a procurement standpoint, contract language and supplier certification could become more standardized thereby improving efficiency. A sales tool that could help a sales representative average four meetings per day compared to three by digitizing the process and removing wasted time would have an immense impact on a large salesforce. R&D leaders using DBS tools to focus new product development around explicit customer needs can greatly enhance R&D efficiency, etc.

As the number and complexity of operating disparate businesses continued to grow, it became apparent to Danaher's leadership that given the significant differences between the businesses the company owned, management could be simplified and value could be maximized by separating the company into two separate businesses, each with its own shareholder base, capital structure, and capital allocation priorities. The less-cyclical, higher growth, consumables-driven businesses would stay with Danaher, and the legacy industrial assets would be spun out into a new public company, Fortive (FTV). The FTV spinout occurred in 2016 and was followed by another spinout, this time of Danaher's dental assets in 2019 into new company called Envista (NVST). The company's most recent spinout occurred in September of last year when it separated its Environmental and Applied Solutions segment into another publicly traded company called Veralto. This segment includes companies in the water testing and treatment industry and product identification business which includes printing technology for the pharmaceutical and packaged foods industries. With the completion of the spin, Danaher is completely focused on the life sciences sector.

I think Danaher views the life sciences industry as attractive because it checks all the boxes: high gross margin, low cyclicality (COVID not withstanding), secular growth, high recurring revenue, low capital requirements and compatibility with DBS. The next section of this writeup will provide more details about the individual businesses and Danaher's financials and valuation.

Business Overview




Diagnostics

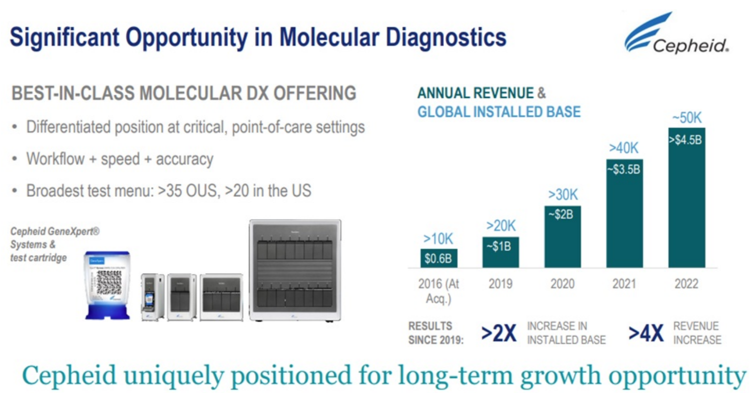
Danaher's largest division is Diagnostics, followed by Biotechnology and Life Sciences. Danaher's diagnostics businesses generated $9.6 billion in revenue in 2023 and include:

- Cepheid, which was acquired in 2016 for approximately $4 billion, is the global leader in molecular diagnostics. There are various types of molecular diagnostic tests including tests for critical infectious diseases including COVID/Flu/RSV/Strep, oncology and genetics, healthcare associated infections such as MRSA, and STDs. As Cepheid was one of the first companies to develop a molecular test for COVID-19, it experienced tremendous growth during the pandemic. Its installed base doubled from the beginning of 2020 to the end of 2022, with over 50,000 testing instruments installed in healthcare settings worldwide at that time. While COVID testing revenue declined sharply in 2023, the non-COVID testing menu continues to grow as Cepheid rolls out new tests. From the DHR CEO at the JP Morgan Healthcare Conference in January 2024:

Well, so first of all, we do think and see that we continue to take share. We've more than doubled our installed base of GeneXperts, since the beginning of the pandemic and we continue to see consolidation of testing platforms at the point-of-care in hospitals. And really, the pandemic was quite an infomercial for Cepheid because our hospital technicians and doctors became so accustomed to getting the right answer at the right time is such an easy workflow. So we continue to see share gain there, and we expect that to continue, and we continue to place more instruments. Now as it relates to the menu expansion, we're seeing that in 2 ways: one, in the

- Danaher acquired Beckman Coulter in 2011 for $6.8 billion dollars. Beckman produces products used in disease diagnosis as well as various life science tools including centrifuges, cell counters, immunoassay analyzers, and flow cytometers. Beckman's annual revenue was $3.7 billion at the time of acquisition. Beckman's products are used in multiple healthcare settings and laboratories. Since the acquisition, Danaher has accelerated the development of new products and increased the organic growth rate of the business from LSD to MSD.
- Leica Biosystems, acquired in 2005, was Danaher's first investment in the life sciences sector. The company is a global leader in high precision optical instrumentation used in microscopy, pathology diagnostics, surgical microscopes, and other applications. Leica's pathology diagnostics business is included in the Diagnostics business as its equipment is used in the examination of tissues and cells to diagnose many diseases, including cancer. Annual revenue was $660 million at the time of acquisition. Its products are used in both academic and pharmaceutical research settings.

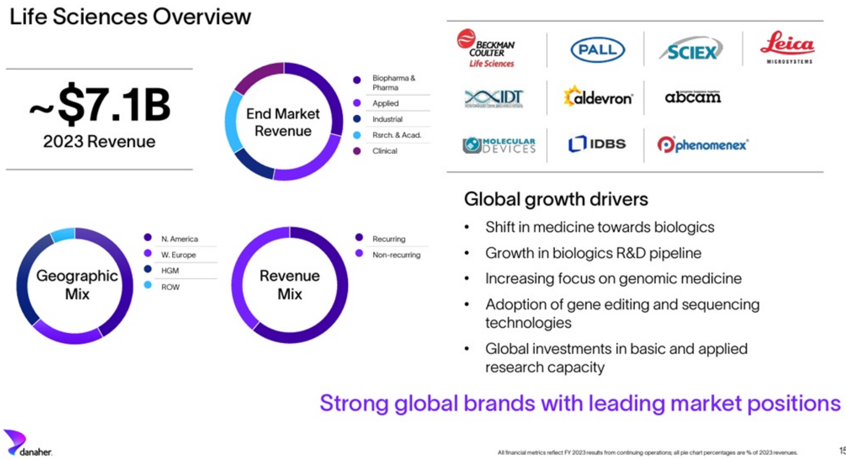
Collectively, Danaher's diagnostics businesses manufacture and sell testing equipment and related consumables to hospitals, physician practices, reference laboratories and other outpatient settings, pathology labs, etc. The sales of consumables are recurring and serve mission critical applications. The company believes that recurring revenue gives DHR more touch points, customer intimacy, and insights into the pain points and the unmet needs of customers which fuels innovation for growth and gaining additional market share. In addition, once a system is installed and integrated into a healthcare provider's workflows, it can be very difficult to displace.




Life Sciences

The Life Sciences division generated $7.1 billion in revenue in 2024 and includes a vast array of highly technical instruments and consumables that are used by researchers to study the basic building blocks of life including DNA, RNA, nucleic acid, proteins, metabolites, and cells to understand the causes of disease, identify new therapies, and test and manufacture new drugs, vaccines, and gene editing technologies. These products are used in initial research which may lead to discoveries of commercially viable medicines which may eventually be produced using products and services from Danaher's Biotechnology division. Primary life science product categories include:

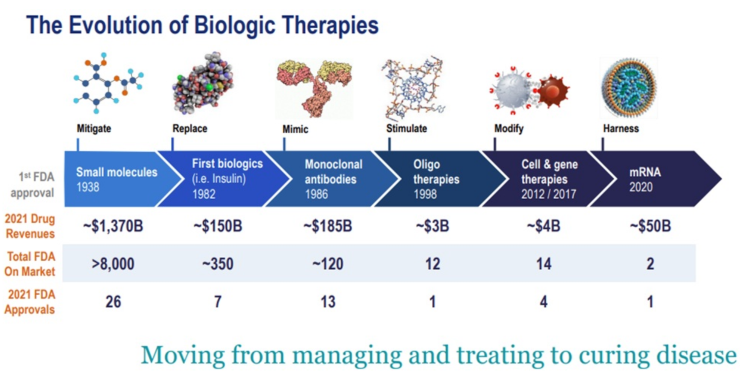
- Flow cytometry, genomics, lab automation, centrifugation, and particle counting – Through its various subsidiaries, Danaher sells instruments and consumables which help researchers study genomic, protein, and cellular information. Researchers use these products to study biological function as well as therapeutic and diagnostic development. Typical customers include pharmaceutical and biotechnology companies, universities, medical schools and research institutions.
- Mass spectrometry – Through its SCIEX subsidiary, Danaher is a global leader in mass spectrometry, which is a technique for identifying, analyzing, and quantifying elements, chemical compounds, and biological molecules, individually or in complex mixtures. Generally speaking, if you want to know what something is made of, a mass spectrometer is the tool you would use. Within life sciences, mass spectrometers are used in numerous applications such as drug discovery and clinical development of therapeutics as well as in basic research, clinical testing, food and beverage quality testing and environmental testing. Typical users of these mass spectrometry and related products include molecular biologists, bioanalytical chemists, toxicologists, and forensic scientists as well as quality assurance and quality control technicians.
- Microscopy—The microscopy business, through brands such as Leica Microsystems, is a leading global provider of professional microscopes designed to capture, manipulate, and preserve images and enhance the user's visualization and analysis of microscopic structures. The Company's microscopy products include laser scanning (confocal) microscopes, compound microscopes and related equipment and software, surgical and other stereo microscopes, and specimen preparation products for electron microscopy. Typical users of these products include research, medical and surgical professionals operating in research and pathology laboratories, academic settings, and surgical theaters.
- Genomic consumables—The genomic consumables businesses, comprised of the 2021 $9.5 billion acquisition of Aldevron, the 2018 $2.1 billion acquisition of Integrated DNA Technologies (IDT) are leading providers of custom nucleic acid products for the life sciences industry, primarily through the manufacture of custom DNA and RNA oligonucleotides and gene fragments utilizing a proprietary manufacturing ecosystem. The businesses have developed proprietary technologies for genomics applications such as next generation sequencing, CRISPR genome editing, qPCR, and RNA interference. Additionally, the businesses are a leading manufacturer of high-quality plasmid DNA, RNA, and proteins which are used in the research, development and manufacture of gene and cell therapies, DNA and RNA vaccines and gene editing technologies. While genomic medicine is still in its infancy, this promises to be a rapidly growing area over time. IDT's 2017 revenue was $260 million and Aldevron's 2020 revenue totaled $300 million which compares to over $1 billion in genomic revenue combined in 2022. Typical users of these products include scientists in the areas of academic and commercial research, agriculture, medical diagnostics, pharmaceutical development, biotechnology companies and research institutions across discovery, clinical and commercial applications.



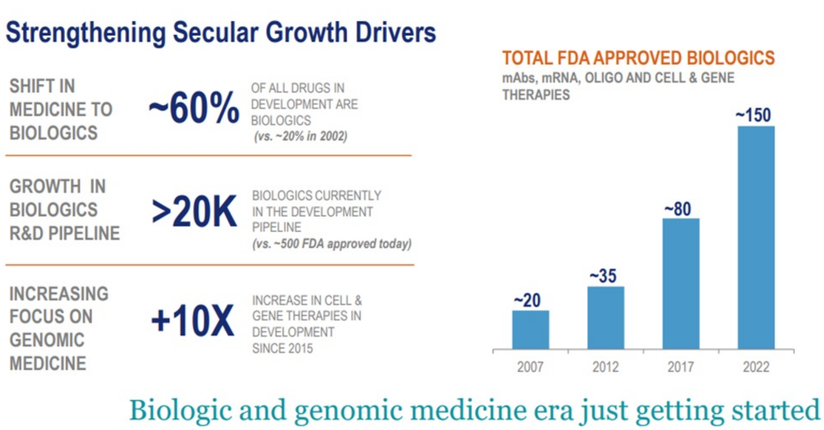

Biotechnology

Like Diagnostics, Danaher has assembled one of the largest providers of bioprocessing products and services to the biotechnology/pharmaceutical industry. The bioprocessing industry produces biologic and genomic based medicines, which are produced from living cells and work by targeting specific cells or proteins in the body. Biologics include replacement therapies such as insulin, vaccines, recombinant proteins and other biologic drugs, to novel cell, gene, mRNA and other nucleic acid therapies. These types of therapies are all “large molecules,” require bioprocessing to manufacture, and are different from “small molecule” traditional drugs which are primarily produced by synthesizing various active pharmaceutical ingredients. The largest biologic on the market currently is AbbVie’s Humira.

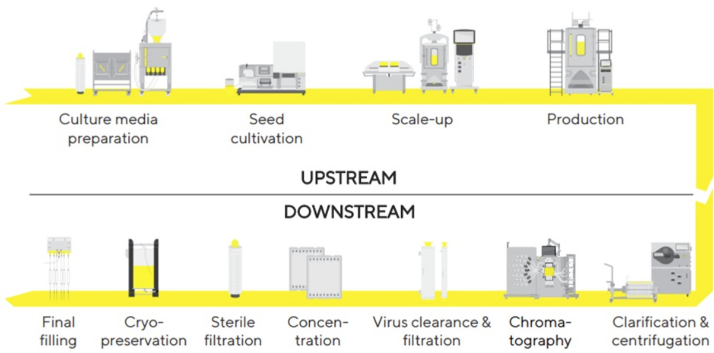
The following slides from the Cytiva investor day in 2022 reflect how the size of the biologics market is a fraction of the overall pharmaceutical industry. Biologic therapies comprised only 22% of the total 2021 pharmaceutical market by revenues and only 6% by number of therapies approved.




Yet, given scientific progress, the pharmaceutical industry is rapidly shifting towards a future in biologics:



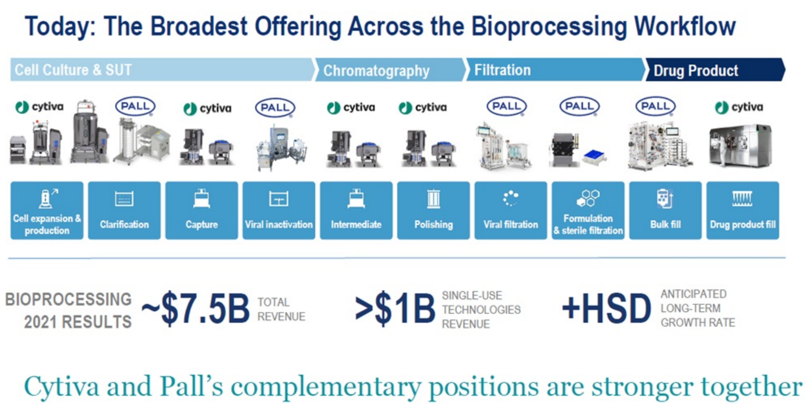

The manufacturing process for biologics is different than for traditional pharmaceuticals. Generally speaking, traditional pharmaceuticals are manufactured in very large batches by mixing active pharmaceutical ingredients and then forming the final product into tablets, pills or liquid solution. (Obviously the process is more complex than this, but it primarily involves mixing ingredients using a few pieces of equipment.) This process typically takes place in stainless steel equipment which must be sterilized before each production run. On the other hand, the production of a biologic is more "delicate", and the typical size of a production run is smaller given the typically smaller applicable patient pool for most biologics (excluding antibody therapies, the COVID vaccine, etc.). Biologics are produced using traditional stainless steel equipment as well as single use technologies (SUT). The process is similar except the equipment (i.e. bioreactors, mixers, filters, etc.) is lined with disposable polymer bags which are discarded after use. The benefits include a reduction in time, labor, water usage, energy, and cleaning chemicals used in the sanitation process. SUT makes small production runs much more efficient and economical. Danaher manufactures both conventional and SUT equipment and consumables. Regardless, whether conventional or SUT, the process is divided into upstream and downstream bioprocessing. From Sartorius' 2022 Annual Report:




The upstream portion of bioprocessing involves the initial identification and culture of specific cells as well as development of the "media" or environment in which the cells will best grow and reproduce. The media is a liquid with nutrients and other things designed to encourage the cells to rapidly multiply. Once the number of cells expands to a certain point (i.e. scale up) they are transferred to a bioreactor for full-scale "production" in which the desired amount of cells is produced. Once the production is complete, there are multiple steps of filtration to remove waste from the desired product as well as purification, concentration, and finish/fill, typically in IV bags or syringes. Danaher has a comprehensive portfolio of products and services in both upstream and downstream bioprocessing.

Danaher's two primary companies in the Biotechnology business are Pall and Cytiva, which have been combined into one operating entity. Pall, which is a leading provider of filtration technologies, was acquired in 2015 for approximately $13.8 billion, which was Danaher's largest ever acquisition at that point. At the time of the deal, Pall generated $2.8 billion in revenue and $663 million of EBITDA. Of the total $2.8 billion in revenue, $1.5 billion came from its life sciences segment and $1.3 came from its industrial segment. In 2022, Pall generated $2.5 billion in life science revenue, with over 80% coming from bioprocessing. Pall's industrial business is currently held within the Life Science segment.

GE Life Sciences (renamed Cytiva) was acquired in early 2020 from General Electric for approximately $20 billion. At the time of the acquisition, Cytiva's revenue and EBITDA were approximately $4.6 billion and $1.2 billion, respectively. Cytiva is a leader in several areas of biologic drug production including bioreactors, cell culture media and chromatography, and it designs and installs full manufacturing suites. The combination of Pall and Cytiva into Danaher's biotechnology group created one of the largest bioprocessing companies in the industry and one which can provide a broad set of offerings across the bioprocessing workflow. These products are sold to biotech companies, pharmaceutical companies, Contract Development and Manufacturing Organizations (CDMOs), Clinical Research Organizations (CROs), academic researchers, etc. CDMOs provide outsourced biologic therapy production services to biotechnology and pharmaceutical companies and CROs provide outsourced clinical trial management.




<u>Competitive Advantage in Biotechnology</u>

Given the relative size and market presence of both Danaher (Pall) and GE Life Sciences (Cytiva), Danaher's acquisition had to receive regulatory approval from the relevant authorities. As part of its review, the EU published a document reviewing the transaction which provided interesting insights into global market share (in 2018) which demonstrated the very strong market position of the two companies. While this information is somewhat dated, it is still relevant for gaining insight into Cytiva's competitive position and why the bioprocessing industry is competitively advantaged, as well as insight into Danaher's post-deal market share in various bioprocessing niches. Specific product categories discussed in the EU's review of the transaction include:

- <u>Bioreactors</u> – In the two primary types of SUT bioreactors, rocking and stirred tank, the combined market share of Danaher and GE was 30% to 40% in both types. This share is nearly all GE as Danaher had only a negligible share prior to the deal. GE's market share in the SUT rocking bioreactor market is second only to Sartorius, with share of 40% to 50%. The stirred tank market is an oligopoly with GE, Sartorius, and Thermo Fisher Scientific (TMO) commanding roughly equal share.

- SUT Mixers – With Danaher and GE both having mixer products, the deal created the largest provider of mixers with a combined share of 30% to 40%, larger than both Sartorius and TMO at 20% to 30% each.
- Microcarriers – Microcarriers are used to provide a surface for cells to attach to from which they can grow in a culture vessel or bioreactor. GE's share in this market was 60% to 70% with Danaher's share of 5% to 10%. The next largest player's share was 10% to 20% for Corning. As a result of the high share, the regulators required the divestment of Danaher's microcarrier business, which still left GE as the largest player by far. Regarding microcarriers, the EU provided the following commentary which speaks to the barriers to entry:

Furthermore, the market investigation suggested that entry of new competitors in this market is not likely. A majority of customers and competitors who expressed an opinion on this point does not expect additional companies to start supplying microcarriers in the next two to three years.[154] Some competitors indicated that the market for microcarriers is characterised by high barriers to entry. In this regard, a competitor expressed the view that "*[t]he extent of expertise, development, and testing required to gain meaningful adoption are high barriers to entry*".[155] The Commission considers that the lack of expected entry of additional competitors would not pose a sufficient competitive constraint on the combined entity post-Transaction.

- SUT connectors provide reliable sterile connections between separate fluid pathways and are used throughout single-use bioprocessing. In the upstream, SUT connectors may be incorporated into bioreactors to enable fluid transfer in or out, ph probe insertion, or sampling. They can also be incorporated into mixer bags. Despite GE and Danaher both having a presence in SUT connectors and a combined market share of 30% to 40%, the strong competitive presence from Millipore Sigma and Sartorius provided sufficient competition and required no remedy. The EU document stated the following regarding high barriers to entry, high technical requirements, as well as scaled competitors.

A significant majority of customers who expressed an opinion in the market investigation on this point did not expect additional companies to start supplying SUT connectors in the next two to three years.[112] A number of these customers pointed towards the existence of high barriers to entry, as newcomers are required to meet the quality and business requirements of mature pharmaceutical companies. In particular, one customer indicated that "*[c]osts and scale of suppliers must be such that new companies coming to the market may not be easy to come by*".[113] The view of competitors in this regard was mixed, with some competitors referring to this market as being "*already occupied by clear market leaders, high technical and quality barriers to entry*" and that, as such, "*[a]ny new entrant would only be able to capture a very small market share*".[114]

- Two of the primary methods of filtration in bioprocessing are direct flow filtration (DFF) and tangential flow filtration (TFF). In DFF, fluids are passed through a membrane. In TFF, the process fluid flows alongside the surface of the filter. The type of filtration used depends on the step of production and the specific application. Both GE and Danaher are active in DFF but they have minimal overlap. Within TFF, there are two primary types of consumable filters, hollow-fiber and flat sheet. In the TFF hollow fiber conventional (i.e. stainless steel) market, the EU estimated that GE and Danaher's combined share at 70% to 80%. In TFF flat sheet SUT, the combined share was estimated at 50% to 60% and in TFF SUT overall, combined share was estimated at 30% to 40%. As a result, the regulators forced Danaher to sell Pall's conventional (i.e. stainless steel) hollow-fiber TFF systems business leaving only GE's market share of 20% to 30%. In addition, the regulators required the sale of Pall's SUT TFF systems business. However, in overall SUT TFF and flat sheet SUT TFF, the Pall business is larger than the GE business so it is unclear what Danaher's market share is post-transaction in these filtration segments. Regarding filtration, the EU provided the following commentary regarding barriers to entry as well as process validation costs:

As indicated in section 4.5.2.1(C.ii), the Commission finds that there are barriers to entry, namely reputation, experience, and the need to offer associated consumables. In addition, entry would only be incentivised when a producer would have an associated consumable in its offering. In that regard, the Commission does not consider it likely that a price increase in systems would trigger entry and to that extent would sufficiently constrain the merged entity. In any event, the Commission does not find serious doubts in the overall market for TFF systems, regardless of the materialisation of this claim.

The Commission however considers it unlikely that customers' buyer power would be able to countervail competition concerns brought forth by the Transaction. In particular, the Commission notes that customers typically buy several products from suppliers in the bioprocessing industry, and are for any of these products that are specified in a commercial scale production line locked in to their supplier as switching would require revalidation and often re-filing with the relevant regulatory approval bodies.[245]

- Chromatography is the core process for purifying the cell mass created in upstream bioprocessing. Chromatography refers to an intricate method of separating and analyzing a complex mixture into its constituent components. There are several different types of chromatography but the one area where Pall and GE overlaped was low-pressure liquid chromatography (LPLC). There are three major components involved in LPLC chromatography: the skid which is essentially the hardware component, the column which is a container which holds resins and is where the process takes place, and the resins themselves, which are complex chemical products which are placed in the column to bind and capture designated proteins. Resins are the most critical component of the process. GE is the clear leader in chromatography skids and resins with Pall having a significantly smaller share. The following chart shows market share for chromatography resins as of 2018:

| | | Danaher | GE | Combined |
|---|---|---|---|---|
| Protein A resins (affinity resins) | Worldwide | [0-5]% | [50-60]-[60-70]% | **[50-60]-[60-70]%** |
| | EEA | [0-5]% | [50-60]-[70-80]% | **[50-60]-[70-80]%** |
| Non Protein A resins (affinity resins) | Worldwide | [0-5]% | [60-70]-[70-80]% | **[60-70]-[70-80]%** |
| | EEA | [5-10]% | [60-70]-[70-80]% | **[70-80]-[80-90]%** |
| Ion exchange resins | Worldwide | [0-5]% | [50-60]-[60-70]% | **[50-60]-[60-70]%** |
| | EEA | [0-5]% | [50-60]-[60-70]% | **[50-60]-[60-70]%** |
| Mixed mode resins | Worldwide | [5-10]% | [50-60]-[60-70]% | **[60-70]-[70-80]%** |
| | EEA | [10-20]% | [50-60]% | **[60-70]-[70-80]%** |

Regarding LPLC standard chromatography skids, GE's share is 60% to 70% for pilot-scale and 50% to 60% for process scale. Pall's respective shares are 10% to 20% and 20% to 30% in these markets, making it a clear number two competitor to GE. For SUT LPLC chromatography skids, GE's share is 70% to 80%. The remedy to this concentration was for Danaher to sell Pall's standard and SUT LPLC chromatography businesses. GE will still be the largest player in these markets.

Regarding resins, Pall sold its resin business although it was very small and only a reseller in the marketplace. Regarding the LPLC continuous chromatography market, Danaher and GE had a combined share of 40% to 50% with GE the smaller player with contributing share of 10 to 20%. The regulators forced Danaher to include Pall's continuous chromatography with the other Pall divestments. In the chromatography conventional column market, combined share is 40% to 50% with Pall contributing 5% to 10% of that. The Pall column business was also divested. Overall, Danaher/GE will still be the leader in LPLC chromatography skids, columns, and resins in both conventional and SUT. Regarding competition and barriers to entry, the EU document mentioned:

> Finally, respondents to the market investigation do not expect additional companies to start supplying chromatography skids (either stainless steel or SUT) in the next two to three years, because of costs, technologies and entry barriers such as regulatory timing.[277]

> Moreover, respondents to the market investigation do not expect additional companies to start supplying chromatography columns in the next two to three years, because, as explained by one customer, "*[i]t takes a long time to get into market and create supply channels etc. so unlikely to be new entrants in timeline above*".[285]

One of the most interesting statements from the EU document describes how the strong market positions within bioprocessing of both Sartorius and GE are due largely to acquisitions that took place over 15 years ago

> In addition, the replies from the market investigation confirmed the Notifying Party's argument that GE's and Sartorius' strong position is due to the fact that they respectively acquired Wave Biotech LLC in 2007 and Wave Biotech AG in 2008, which resulted from the split of Wave Biotech, the inventor of SUT rocking bioreactors, into two companies.[46] In this regard, a customer elaborated as to GE's and Sartorius' relative position in a market for SUT rocking bioreactors: "*[t]he rocking bioreactor technology was developed by Wave Biotech over 20 years ago and subsequently acquired by GE. A European based copy of the same technology was purchased by Sartorius. These have been the dominant players in this market place*".[47] This is consistent with the replies from competitors, who stated that "*[s]trongest players were first to market with their SUT bioreactors*" and that "*Thermo has had a long-standing leadership in classical single-use bioreactors*".[48]

Wave Biotech was the company that invented SUT bioreactors for the bioprocessing industry. After the company split itself between its US and European operations, GE purchased the US company in 2007 when the company's revenue was $25 million. The following year Sartorius purchased the slightly smaller European business (revenues of 10 million Euro). The fact that Sartorius and Cytiva (i.e. GE) are two of the largest suppliers to the bioprocessing industry speaks to barriers to entry within the industry as well as the difficulty of getting customers to switch suppliers. One reason for this is that as part of a regulatory review for the approval of a new biologic drug, the FDA must approve not only the drug, but also validate the production process including the equipment, down to the supplier. Once a drug is filed with the FDA (which is obviously prior to large scale commercial production) it's rare to switch suppliers as that would require an expensive and time-consuming re-validation. As a result, this creates a lock-in effect (very much like the aerospace OEM aftermarket dynamic). This also means that the initial product sales which begin in the R&D stages are relatively small and will only become lucrative once/if the drug succeeds in clinical trials, is approved by the FDA, and is adopted in sufficient quantity by the market. In other words, the addressable market is quite small (relative to the overall market) for a new entrant attempting to gain traction in the industry. For example:

> As to supply-side substitutability between different types of bioprocessing filtration consumables, the results of the market investigation do not contradict this. One competitor for instance indicates: "*[t]he barriers to introducing a new TFF cassette are mostly a long ROI, as swapping a cassette in an existing process would be very difficult, so you would need to specific at Clinical phase 1 and grow with the process. This could take 5+ years before you are in manufacturing and seeing any ROI. In the meantime you would incur all the costs of manufacturing supporting and validating the device*".[172]

Again, this dynamic is like the aerospace sales cycle in which the OEM component providers make little money in the first few years when only selling to Boeing/Airbus, yet make up for it on the back end with high-margin aftermarket sales which can last for many years. Also, like aerospace aftermarket sales, SUT liners, tubes, and filters as well as consumables for conventional production result in a steady stream of high margin “aftermarket” revenue so long as the drug is produced.

An additional factor which limits new entrants into the marketplace is simply the technological sophistication of the equipment and the reputational importance of suppliers. A biotech or pharmaceutical company won’t want to partner with a supplier who may not be in business in a decade or who is unable to maintain the highest standards of quality and customer service. A corollary to this is that for producers of biologic drugs, the production costs associated with bioprocessing are a relatively small amount compared to the overall costs of R&D, clinical trials, regulatory approval, marketing, distribution, etc. As such, industry suppliers compete less on cost and more on technological expertise and customer service. These factors are supported by the fact that there are only a handful of large suppliers to the industry, including Cytiva, Sartorius, Thermo Fisher, and Merk Millipore among others, which have dominant market shares in the various niches.

Financial Overview

The following chart shows Danaher’s revenue and operating profit by segment. For years prior to 2020, Biotechnology is included in Life Sciences. Core growth is defined as reported revenue adjusted for acquisitions and divestitures and foreign currency movement.

| ($ in millions) | Actual | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 |
| Revenue: | | | | | | | | |
| Biotechnology | | | | | 5,276 | 8,570 | 8,758 | 7,172 |
| *Core growth* | | | | | | *29.5%* | *6.0%* | *-18.0%* |
| Life Sciences | 5,366 | 5,710 | 6,471 | 6,951 | 5,300 | 6,388 | 7,036 | 7,141 |
| *Core growth* | *3.5%* | *4.0%* | *7.5%* | *7.0%* | *13.0%* | *17.0%* | *9.5%* | *1.0%* |
| Diagnostics | 5,038 | 5,840 | 6,258 | 6,561 | 7,403 | 9,844 | 10,849 | 9,577 |
| *Core growth* | *2.5%* | *4.0%* | *6.5%* | *7.0%* | *13.5%* | *31.0%* | *13.5%* | *-10.5%* |
| Total | 10,404 | 11,550 | 12,729 | 13,512 | 17,979 | 24,802 | 26,643 | 23,890 |
| *Core growth* | *3.0%* | *5.5%* | *6.0%* | *6.0%* | *9.5%* | *25.0%* | *9.5%* | *-10.0%* |

| | Actual | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 |
| Operating profit: | | | | | | | | |
| Biotechnology | | | | | 1,082 | 3,074 | 3,008 | 1,909 |
| *Margin* | | | | | *20.5%* | *35.9%* | *34.3%* | *26.6%* |
| Life Sciences | 819 | 1,004 | 1,229 | 1,401 | 972 | 1,293 | 1,414 | 1,209 |
| *Margin* | *15.3%* | *17.6%* | *19.0%* | *20.2%* | *18.3%* | *20.2%* | *20.1%* | *16.9%* |
| Diagnostics | 786 | 872 | 1,074 | 1,134 | 1,538 | 2,313 | 3,436 | 2,406 |
| *Margin* | *15.6%* | *14.9%* | *17.2%* | *17.3%* | *20.8%* | *23.5%* | *31.7%* | *25.1%* |
| Operating profit: | 1,605 | 1,876 | 2,303 | 2,535 | 3,592 | 6,680 | 7,858 | 5,524 |
| *Margin* | *15.4%* | *16.2%* | *18.1%* | *18.8%* | *20.0%* | *26.9%* | *29.5%* | *23.1%* |
| Corporate costs | - | - | - | - | - | 303 | 322 | 322 |
| Net operating profit | 1,605 | 1,876 | 2,303 | 2,535 | 3,592 | 6,377 | 7,536 | 5,202 |

Prior to the pandemic, the businesses generated steady mid to high-single digit organic growth, but the pandemic supercharged growth in diagnostic testing and bioprocessing. For Diagnostics, overall organic revenue growth averaged 5% in the four years leading up to 2020 and margins have expanded over time. Total revenue in 2022 was $10.8 billion, including a significant bump from COVID testing. For 2023, revenue related to respiratory testing (a significant amount related to COVID) declined by over $2 billion. However, with LDD growth in Cepheid’s non-respiratory testing revenue and growth in other areas of the business, overall revenue only declined to around $9.5 billion, or a decline of 11%. Margins declined in 2023 due to lower testing volume in addition to costs related to capacity and labor reductions at facilities dedicated to COVID the past two years. Over the next several years, the company anticipates Diagnostic revenue growth in the HSD range over the next few years given LDD anticipated growth in molecular testing (Cepheid) and HSD growth in pathology and acute care.

The operating margin has risen from 15.4% in 2015 to 25.1% in 2023 as the business has experienced operating leverage and increased growth in molecular testing over that time, despite the decline in 2023. Since 2015, revenue has doubled and the operating profit margin has increased 1,000 bps, which has led to operating profits increasing over 3x, from $746 million to $2.4 billion. This margin expansion has benefited from growth in the number of Cepheid molecular testing systems in recent years. For example, the number of testing machines has increased from approximately 20,000 in 2019 to currently over 55,000 worldwide. As the number of testing machines has grown, the numbers of available tests the machines are capable of processing is also increasing. This tremendous growth has led to a significant increase in the amount of testing consumables sold, which generates very high incremental margins. Approximately 80% of Diagnostic revenue is recurring based on a high level of consumable sales and service contracts.

Danaher only broke out the Biotechnology segment in 2022 and provided three years of historical numbers. Biotechnology played a role in production of both COVID vaccines and monoclonal antibody therapies which benefitted revenue in 2021 and 2022. COVID related revenue in 2022 was $800 million (just less than 10% of the total) and declined sharply in 2023. Over the longer-term, Danaher anticipates Biotechnology can grow at a HSD rate given the continued growth in biologics over time. Biotechnology is the company’s highest margins segment given high margin SUT as well as other consumables. Margins rose sharply in 2021 as the business experienced a margin bump during COVID as operating leverage kicked in. However, given the tremendous growth during the pandemic coupled with the slowdown in the funding environment for the biotech space (i.e. 20% to 30% of revenue comes from emerging biotech), and a significant inventory overhang due to overordering during the pandemic, organic revenue growth declined by 10% in 2023. Biotechnology revenue is anticipated to decline by mid-teens rates during the first half of the year before returning to year over year growth in the back half against much easier comparables, resulting in a LSD full-year decline. Danaher believes the worst of the bioprocessing slump is behind it at this point.

The Life Sciences segment averaged 5.5% organic revenue growth in the four years leading up to the pandemic and has averaged double digit growth during the pandemic before slowing to 1% in 2023. The company anticipates MSD growth from Life Science instruments and DD growth from the $1 billion genomics business resulting in HSD organic growth from 2024 and beyond. The company has experienced margin expansion in recent years, although not to the same extent as the other businesses because Life Sciences did not experience a COVID bump to the same extent.

Danaher maintains an investment grade balance sheet with net leverage at 1.4x. As of March 31, cash totaled $7 billion compared to debt of $18.2 billion and 2024 consensus EBITDA of $7.73 billion. The company's debt is termed out with no significant maturity in any given year relative to the generation of free cash flow. The largest single-year maturity is $2.3 billion in 2026 compared to ongoing free cash flow of nearly three times that amount.

While 2023 was and 2024 will continue to be somewhat of a reset given the COVID wind down and inventory destocking in the bioprocessing space, the company continues to anticipate HSD long-term organic revenue growth with strong incremental operating margins (35% to 40%) given the high level of consumable revenue. In addition, given the fact that Danaher doesn't repurchase stock and only pays a small dividend, future cash flows will likely be allocated to acquisitions, which Danaher has used to fuel its growth since the company was founded. Organic and inorganic growth coupled with margin leverage will lead to DD compound growth in earnings and free cash flow per share. Historically the acquisition program has generated strong returns on capital. For example, looking at the incremental growth in EBITDA over the past five and three years, incremental returns on the capital invested in capital expenditures and acquisitions has been HSD/LDD despite the post-pandemic weakness. While Danaher has paid high prices for some of its acquisitions, overall, the results have been good.

Valuation

Based on a stock price of $250 with 741 million shares outstanding, DHR's market capitalization is approximately $185 billion. With net debt of $11.1 billion, the EV is $196 billion. The consensus estimate for 2024 EBITDA is $7.73 billion, which values the company at just over 25x EV/EBITDA. Free cash flow generated in 2023 totaled to $5.8 billion resulting in a free cash flow yield of 3.1% on a trailing basis. Given the declines in Biotechnology this year, free cash flow will likely be flattish in 2024 and then revert to growth in future years. With a 3% starting free cash flow yield and LDD growth in earnings/free cash flow per share over the intermediate to long-term, I think long-term double digit compounding is likely. I can model HSD compounding in the stock price over the next five years assuming the free cash flow multiple derates to 25x along with HSD revenue growth and a small amount of operating leverage with cash flow recycled into share repurchases.

While the current multiple EV/EBITDA multiple of 25 wouldn't be considered cheap in absolute terms, it is within the stock's trading range over the past five years, despite the fact that the company today has higher margins, faster revenue growth, and greater recurring revenue as compared to periods prior to the Envista and Veralto spinoffs.

Risks

- Given the highly technical areas in which Danaher operates, it faces the risk of technological obsolescence or getting caught by competitors. For example, competitive advancements in certain areas of bioprocessing could lead to share losses for Cytiva. The counter to this is that given the heavy oversight of the FDA, likely changes over time would be incremental and lengthy. Also, Danaher, being one of the largest providers in the areas in which it competes, spends vast sums on R&D and capital expenditures, allowing it to stay on the cutting edge from a technological standpoint. Plus, the company's diversified healthcare business makes the company more robust and not wholly dependent on a certain product or service.
- By only paying a small dividend and not repurchasing stock, Danaher recycles its free cash flow into acquisitions. In the past the company has created significant equity value via its acquisition program and the application of DBS; however, it has been willing to pay more for recent deals as it has moved into higher-growth markets. For example, in 2021 it acquired a genomics company called Aldevron for $9.5 billion despite Aldevron only generating $300 million in 2020, the year prior to being acquired. Aldevron has grown very rapidly, and it generates very high margins, so the deal could work out. However, acquiring smaller, rapidly growing companies for very large premiums makes the deal math that much more difficult, especially with rates having moved higher. Given the track record, I'm willing to give them the benefit of the doubt with the M&A program; however, this will require scrutiny moving forward.
- While Danaher stock is cheap relative to its trading multiples in recent years, it isn't inexpensive on a stand-alone basis. While I think a starting free cash flow yield of 3.1% is attractive given the qualities of the business, the company will have to continue to generate long-term HSD revenue growth going forward for the stock to compound at a DD rate. If growth disappoints, the multiple could de-rate. I think given the growth of the healthcare industry overall, it is not a question of will Danaher grow, but rather by how much. If grow is muted relative to expectations for the next few years, it could take a while for the stock to grow into its valuation.

Summary

With the stock price not having moved much in recent years, I believe the excesses have mostly been worked off and now is a good time to buy Danaher at a trough 3% free cash flow yield. After the Veralto spin, Danaher is now a competitively advantaged, diversified life sciences company providing critical products with a steady stream of high-margin consumable revenue operating in niches with secular growth tailwinds. The company has an excellent long-term track record of acquiring and improving companies using DBS, as evidenced by very strong long-term equity returns. In addition, investors get the opportunity to invest alongside the company's founders, the Rales brothers, who maintain a collective ownership position in the company of over 10%, valued at nearly $19 billion.

I do not hold a position with the issuer such as employment, directorship, or consultancy.
I and/or others I advise hold a material investment in the issuer's securities.

## Catalyst

- Upturn in bioprocessing industry
- Accretive M&A

# Messages

**Subject** Share Repurchases
**Entry** 07/23/2024 07:59 AM
**Member** nassau799

Thanks for a thorough and interesting analysis, Valueinvestor.  I'm confused on one point:  You say twice that DHR does not buy back stock.  Yet in the Valuation section future repurchases are part of your growth algorithm.  Is this tied to a belief that DHR is running out of acquisition targets and will turn to that as an option?

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**Subject** Re: Share Repurchases
**Entry** 07/23/2024 08:14 AM
**Member** valueinvestor03

DHR will not repurchase stock in reality. I just assumed share repurchases for modeling purposes to soak up free cash flow. In all likelihood they will continue to do M&A but I didn't attempt to model this given the inherent uncertainties with timing and magnitude. I think M&A will ultimately produce better returns than share repurchases so I think modeling repurchases produces a conservative estimate of future compounding.

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**Subject** Re: Re: Share Repurchases
**Entry** 07/23/2024 08:17 AM
**Member** coffee1029

just announced $4.5bn buybacks in recent weeks, first since 2012 and largest for at least 20 years

CEO framed them as more attractive than currently actionable M&A, which remains their bias for capital deployment

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**Subject** Re: Re: Re: Share Repurchases
**Entry** 07/23/2024 08:26 AM
**Member** valueinvestor03

They want to do M&A. I'd be surprised if they bought back stock.

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**Subject** Re: Re: Re: Re: Share Repurchases
**Entry** 07/23/2024 08:44 AM
**Member** valueinvestor03

On the call this morning the company stated it has repurchased approximately 19 million shares through Q2 and into Q3. Obviously I didn't expect this....first time they've bought back stock in like a decade.

I'll take it.

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**Subject** Flag for removal?
**Entry** 07/23/2024 11:34 PM
**Member** coffee1029

Thanks for the detailed write up.

It probably took several hundred times the amount of work that another VIC member spent on fully reading it and deciding to tag this idea "flag for removal."

To that member I suggest some commentary is expected.  "Flag for removal" is a useful tag, but I struggle to understand it here.

**What else is expected of VIC Members?**

Members should rate ideas considerately and after fully reading them. If a member rates an idea very poorly or very highly, some commentary is expected by the other members. While there is no need to publicly declare or explain your rating, pointing out some flaws or errors in an idea rated "1" or "2" will help the author of the idea submit better work next time.